

Knowledge Graphs for Healthcare

M Usman
Semantic Web
M.Sc. (Computing)

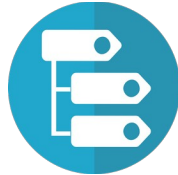


Contents



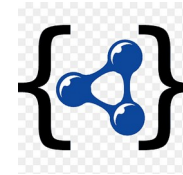
About KG

Intro to KG & Ontologies



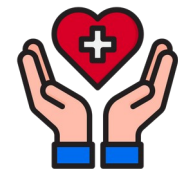
About Healthcare

KG in Healthcare



Applications

General ^ Specific Purpose
Ontologies



Benefits & Way forward

Healthcare Future



About KG

Intro to KG & Ontologies

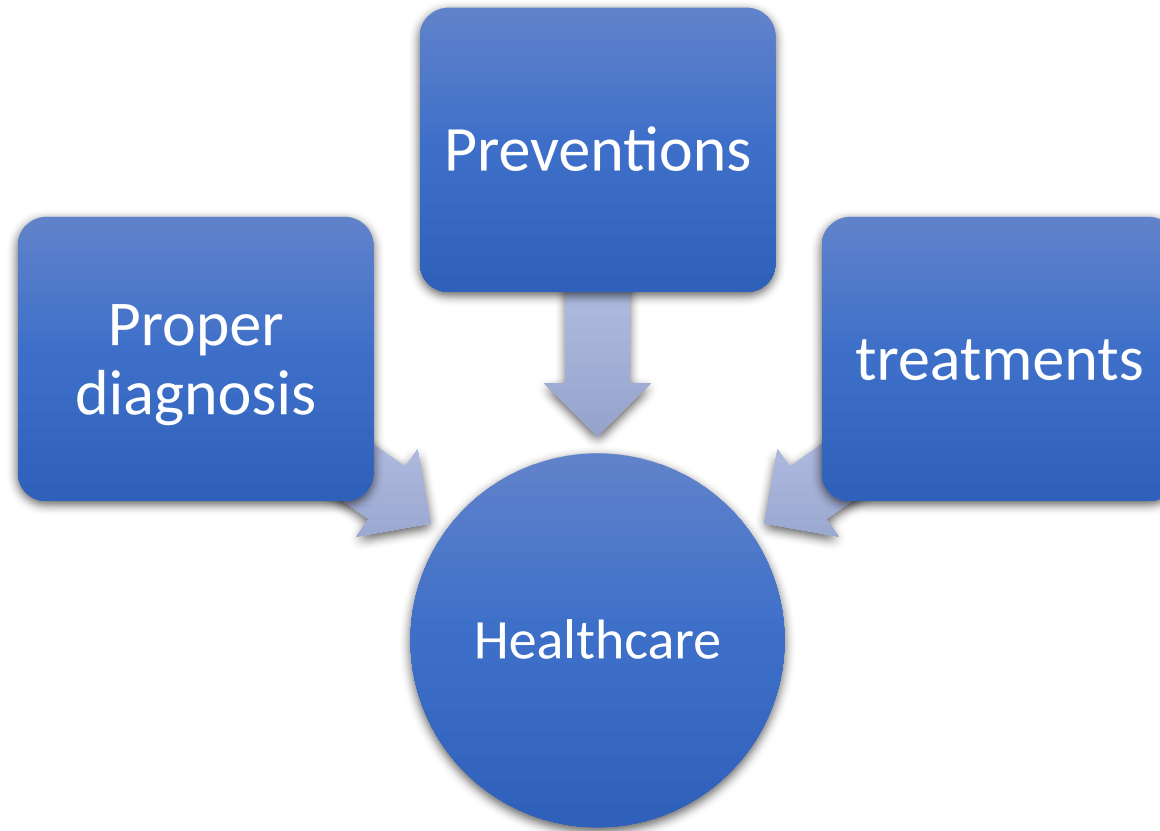
Introduction to Knowledge Graph & Ontologies

KG for Healthcare

Why KG is important in Healthcare



About KG



Challenges:

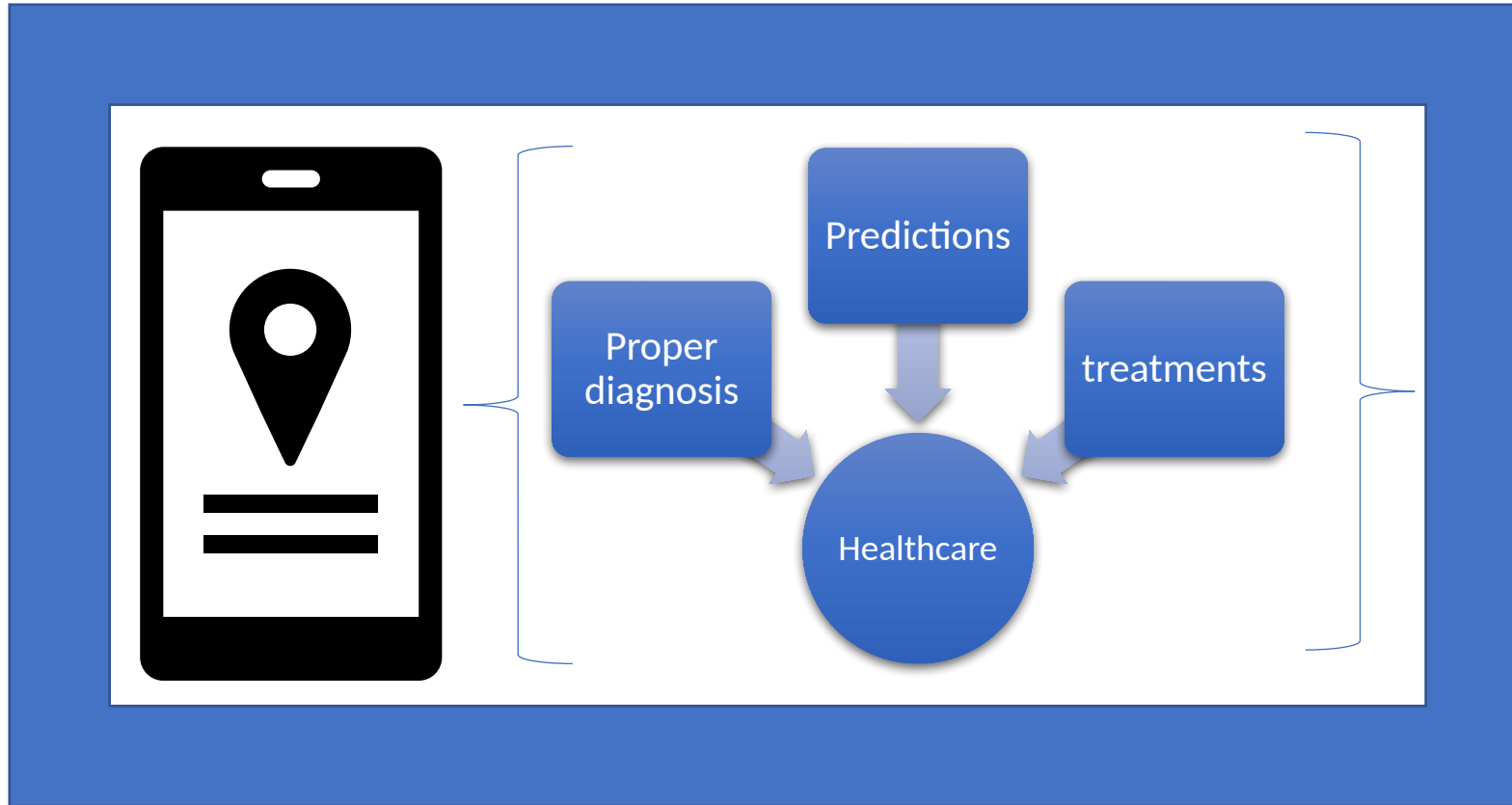
- Time
- Accuracy
- Coverage

Typical healthcare functions

Why KG is important in Healthcare



About KG



Powered by KG

Benefits:

- Time
- Accuracy
- coverage

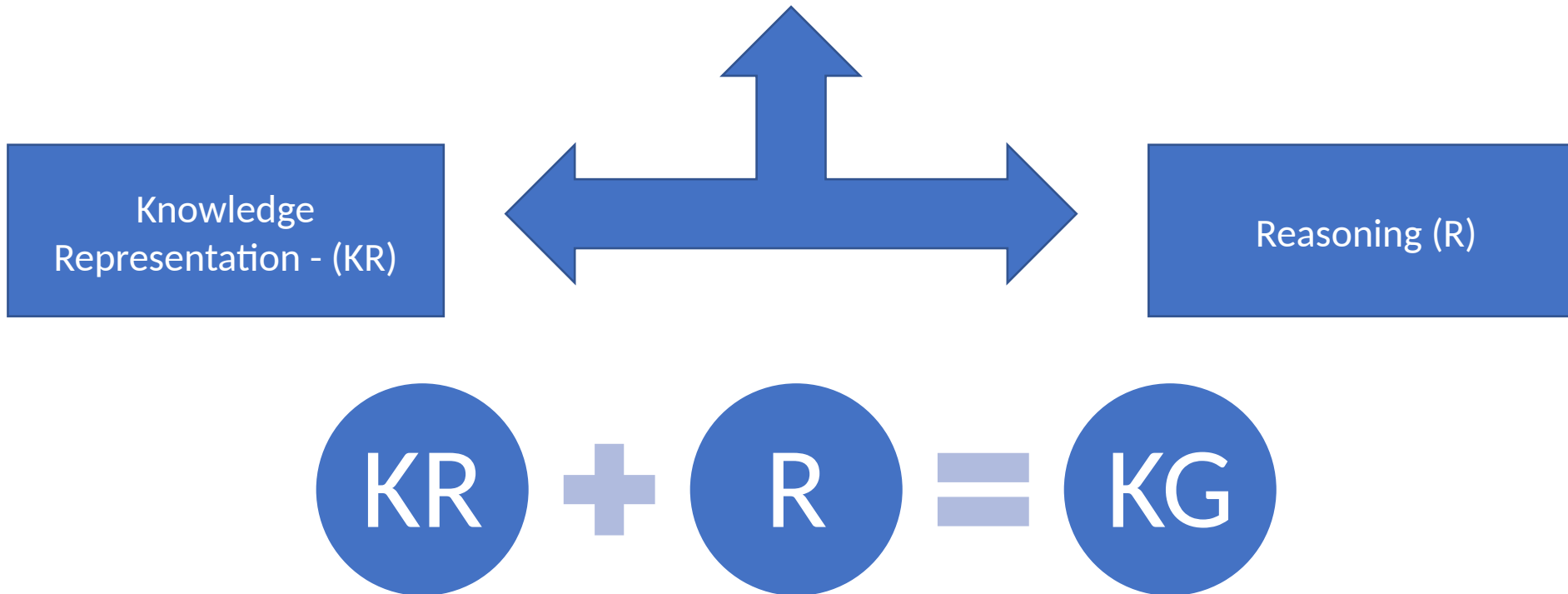
Pervasive healthcare system



About KG

What is knowledge graph - KG?

Knowledge Graph acquires and integrates information into an ontology and applies a reasoner to derive new knowledge.



KG and Ontology



About KG

Knowledge Graph

- Datasets from multiple sources
- Uncover connections not realized before

Ontology

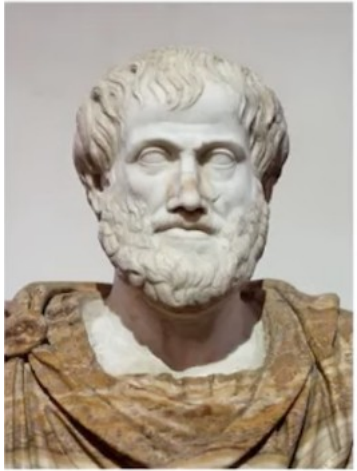
- Organized knowledge base
- Formal representation of entities in the shape of a graph

ontology + data = knowledge graph

Ontology and its Origins



About KG



**Aristotle -
384-322BC**

- First systematic taxonomy of biology
- Classification of organisms based on shared properties



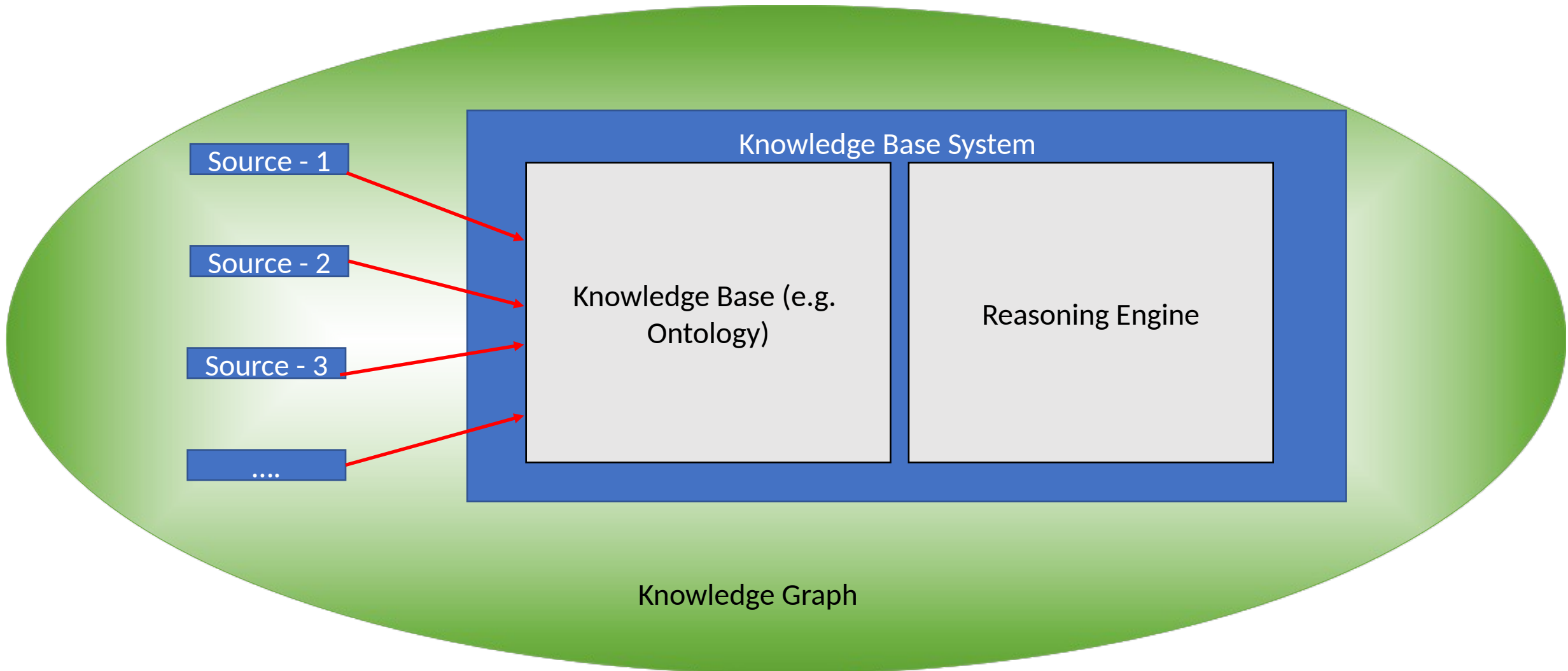
**Galen -
130-210 AD**

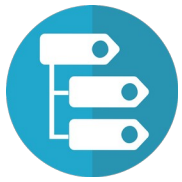
- Systematic description of diseases, signs and symptoms

KG architecture



About KG





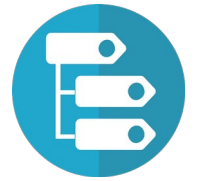
About Healthcare

KG in Healthcare

Introduction to Healthcare Graphs

KG for Healthcare

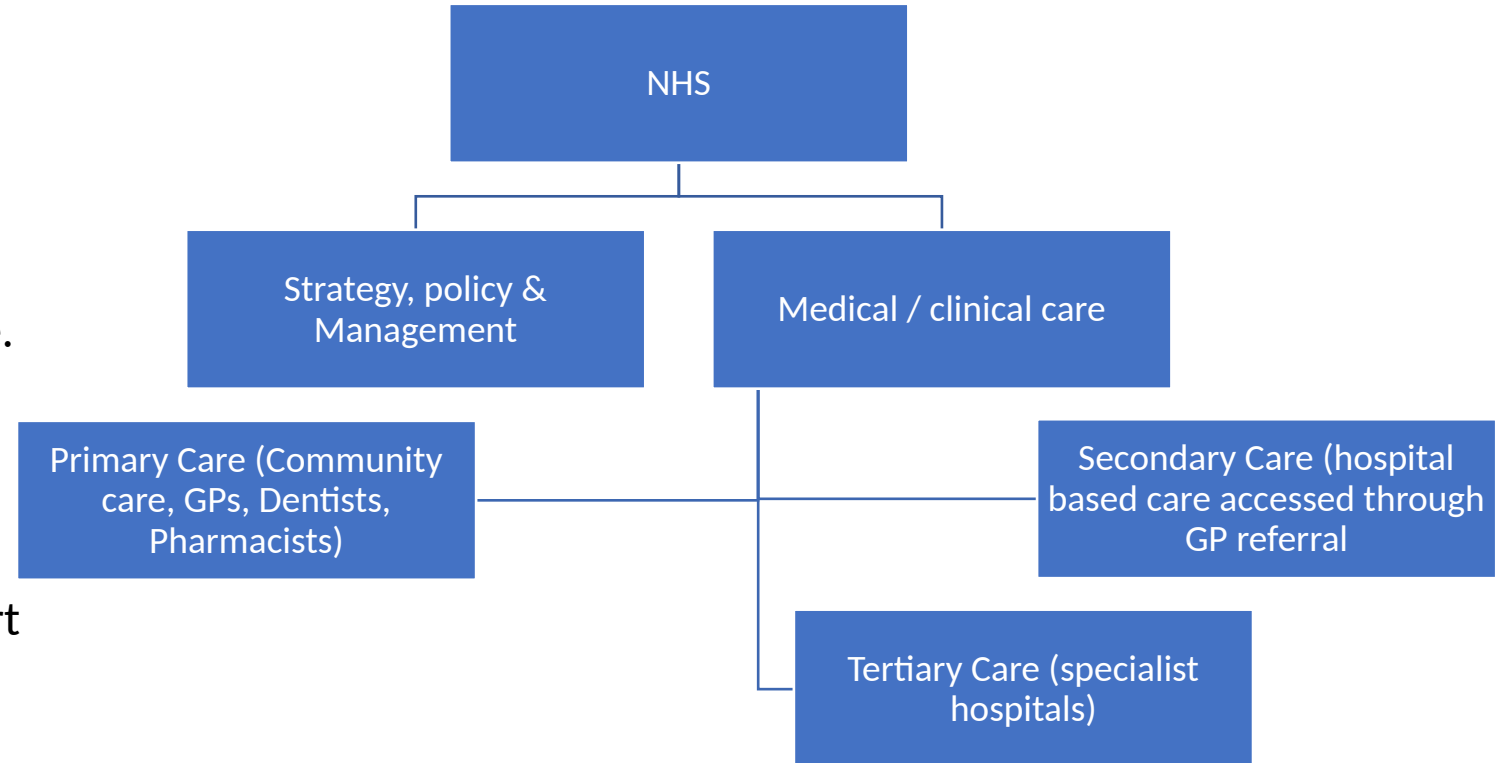
UK HealthCare System



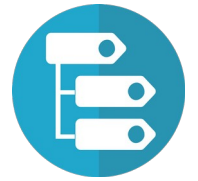
About Healthcare

NHS – Established in 1948 in UK. Similarly, different healthcare systems exist across the world.

- Predictive, Preventive and Personalized Medicine is enabled through telemedicine.
- Preventive Medicine is strong in UK, however predictive and Personalized are becoming so.
- Scientific discoveries play a crucial role in Predictive and personalized medicine apart from public involvement.



Ontologies in Biomedical



About Healthcare

Gene
Ontologies

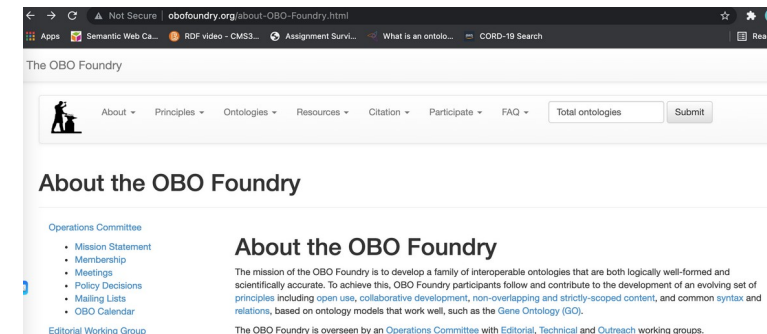
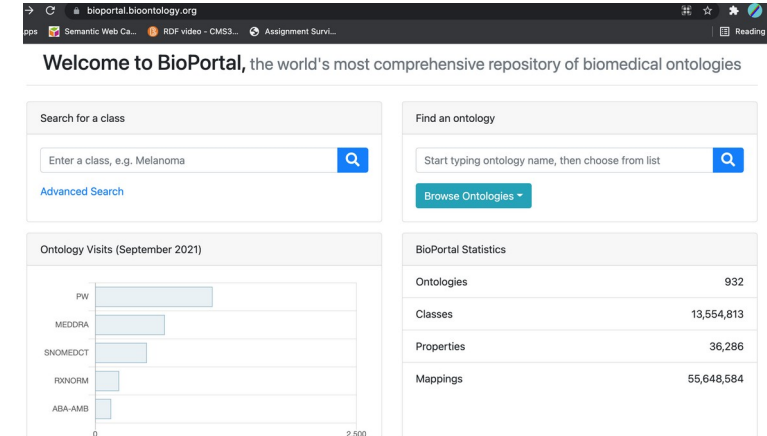
Disease
Ontologies

Phenotype
Ontologies

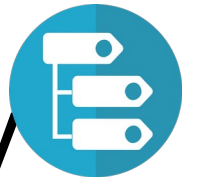
Bio portal has 934 Ontologies as on Sep 2021.

OBO Foundry has 234 Ontologies as on Sep 2021.

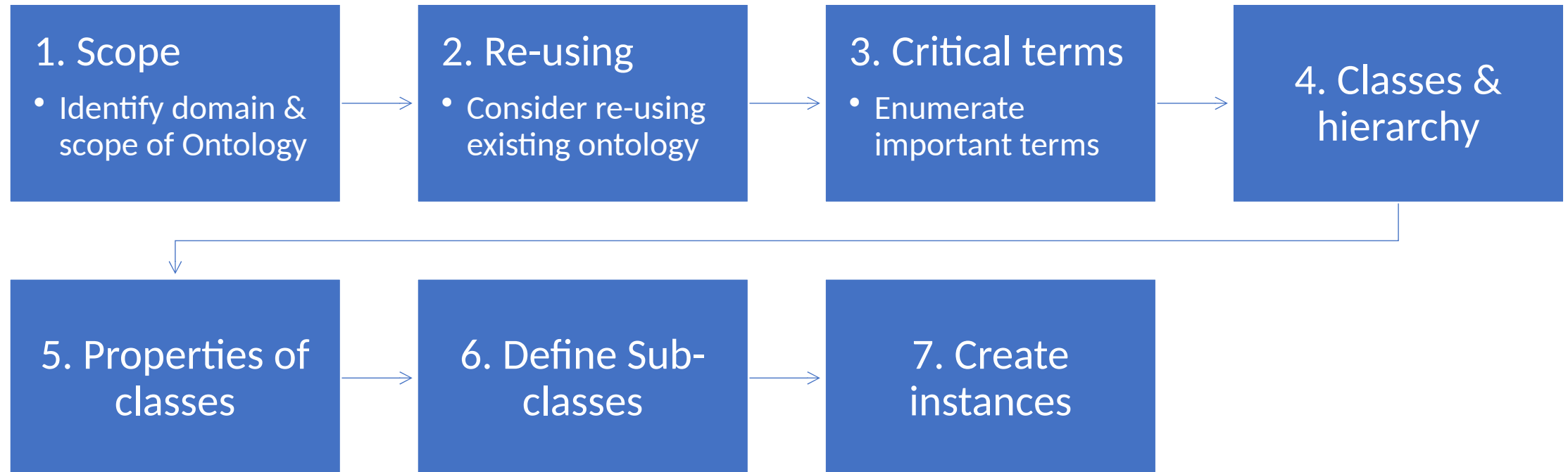
OBO	Ontologies	Class	Object Property	Datatype Property	Annotation Property	Instance	Total
Total	234	5,439,376	17,461	695	14,839	675,679	6,148,050



Steps for creating healthcare ontology



About Healthcare



Ontologies Deployment architecture

<http://www.obofoundry.org/about-OBO-Foundry.html>

<https://bioportal.bioontology.org/>

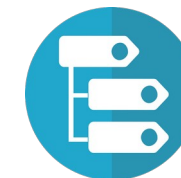
Biomedical
ontologies

Model Mapping

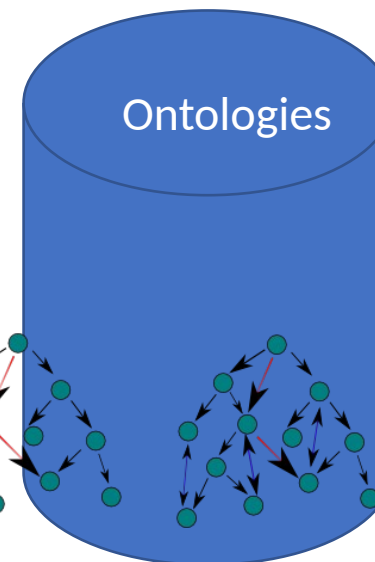
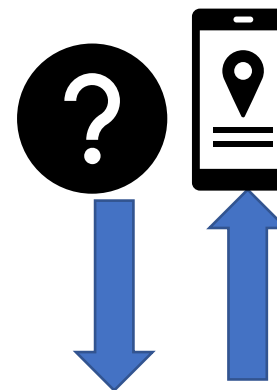
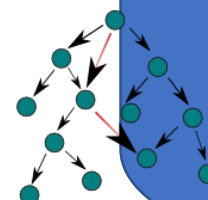
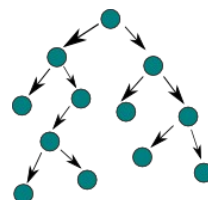
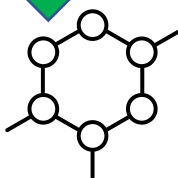
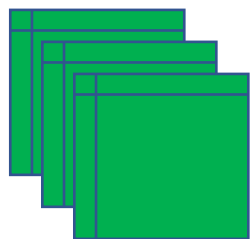
Ontologies

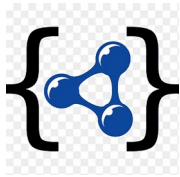
External

Internal



About Healthcare





Applications

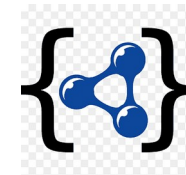
General ^ Specific Purpose
Ontologies

Applications of KG in Healthcare

KG for Healthcare

Use Case: Gene Ontology - GO

General Purpose Ontology



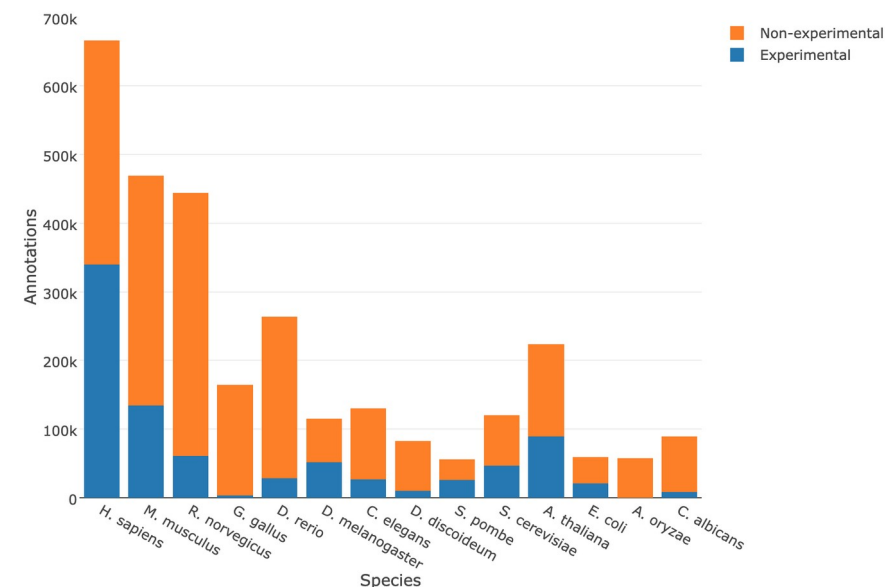
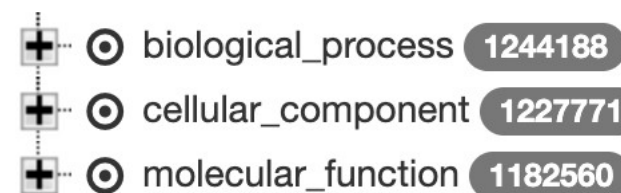
Applications

Go entails “types” to define gene functions into three major functions i.e.

- Biological Process
 - Includes pathways & processes that genes affect upon
- Cellular Component
 - Ascribes parts related to genes
- Molecular Function
 - Identifies composition of genes

GO promotes interoperability of gene functions across multiple species

Total Gene Products: 3 654 519

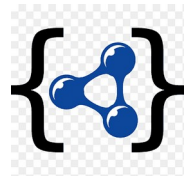


SNOMED CT

General Purpose Ontology

- Controlled coded clinical terminology intended to be used in the Electronic Health Records.
- Developed by UK NHS and College of American Pathologists in US in 2002
- Now maintained by the International Health Terminology Standards Development Organisation (known as IHTSDO).
- In 2017 IHTSDO adopted the trading name SNOMED International.
- SNOMED CT supports information retrieval, DSS, communication and patterns identification.

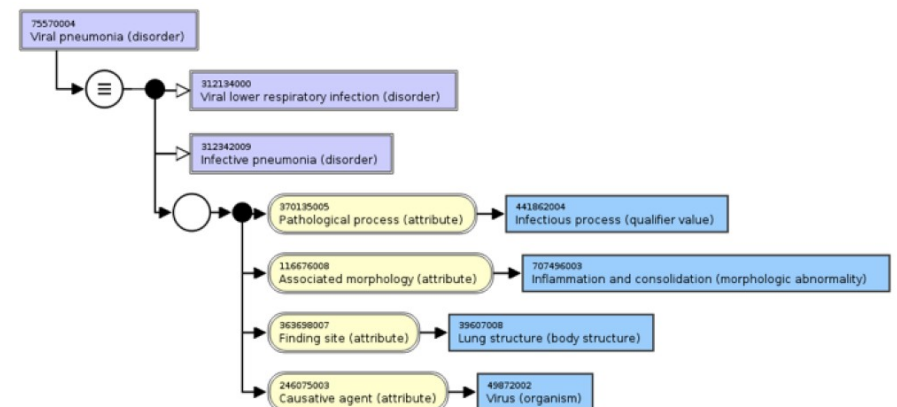
Systematized Nomenclature of Medicine-Clinical Terms (SNOMED-CT)



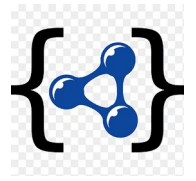
Applications

- Body structure
- Clinical finding
- Environment or geographical location
- Event
- Observable entity
- Organism
- Pharmaceutical / biologic product
- Physical force
- Physical object
- Procedure
- Qualifier value
- Record artifact
- Situation with explicit context
- SNOMED CT Model Component
- Social context
- Special concept
- Specimen
- Staging and scales
- Substance

19 Classes and Sub-Classes > 3 61 569



SNOMED CT



Applications

- One logical framework for representing clinical concepts including diagnosis, signs and symptoms.
- Also includes procedures like surgical, therapeutic, diagnostic and observables like heart rate
- Reduces the need for multiple incompatible or overlapping code systems

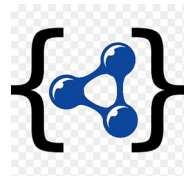
 Argentina	 Denmark	 Kazakhstan	 Singapore
 Armenia	 Estonia	 Lithuania	 Slovakia
 Australia	 Finland	 Luxembourg	 Slovenia
 Austria	 Germany	 Malaysia	 Republic of Korea
 Belgium	 Hong Kong, China	 Malta	 Spain
 Brunei	 Iceland	 The Netherlands	 Sweden
 Canada	 India	 New Zealand	 Switzerland
 Chile	 Ireland	 Norway	 United Kingdom
 Cyprus	 Israel	 Portugal	 United States
 Czech Republic	 Jordan	 Saudi Arabia	 Uruguay

SNOMED CT

Last uploaded: May 20, 2021

Summary		Classes	Properties	Notes	Mappings	Widgets
Details			Metrics ?			
Acronym	SNOMEDCT		Classes	361,588		
Visibility	Public		Individuals	0		
Description	SNOMED Clinical Terms		Properties	242		
Status	Production		Maximum depth	28		
Format	UMLS		Maximum number of children	1,229		
Contact	NLM Customer Service, custserv@nlm.nih.gov		Average number of children	4		
Categories	Health		Classes with a single child	42,973		
Groups	Cancer Biomedical Informatics Grid, Unified Medical Language System		Classes with more than 25 children	2,967		
License Information	This ontology is made available via the UMLS. Users of all UMLS ontologies must abide by the terms of the UMLS license, available at https://uts.nlm.nih.gov/license.html		Classes with no definition	355,600		

These 40 countries are members of SNOMED International.

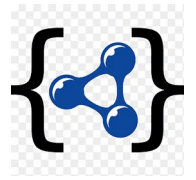


Applications

General Purpose Ontologies

Vocabulary	Purpose
ICD-9-CM/ICD-10 (WHO, 2008)	disease classification, in health statistics, hospital billing
WHO Drug Dictionary (UMC, 2008), ATC (WHOCC, 2008), RxNorm (NLMa, 2008) DM+D (NHS, 2008)	drug classification
NCI Thesaurus and Metathesaurus (NCI, 2008)	cancer research
LOINC (REGENSTRIEF INSTITUTE, 2008)	inter-laboratory communication
MedDRA (NORHTROP GRUMMAN, 2008)	medicine-related regulatory activities
DICOM (MITA, 2008)	medical image and imaging process descriptions
MeSH (NLM, 2008)	medical literature indexing
SNOMED CT (IHTSDO, 2008)	clinical documentation

Specific Purpose Ontology: Covid-19 KG (CKG)

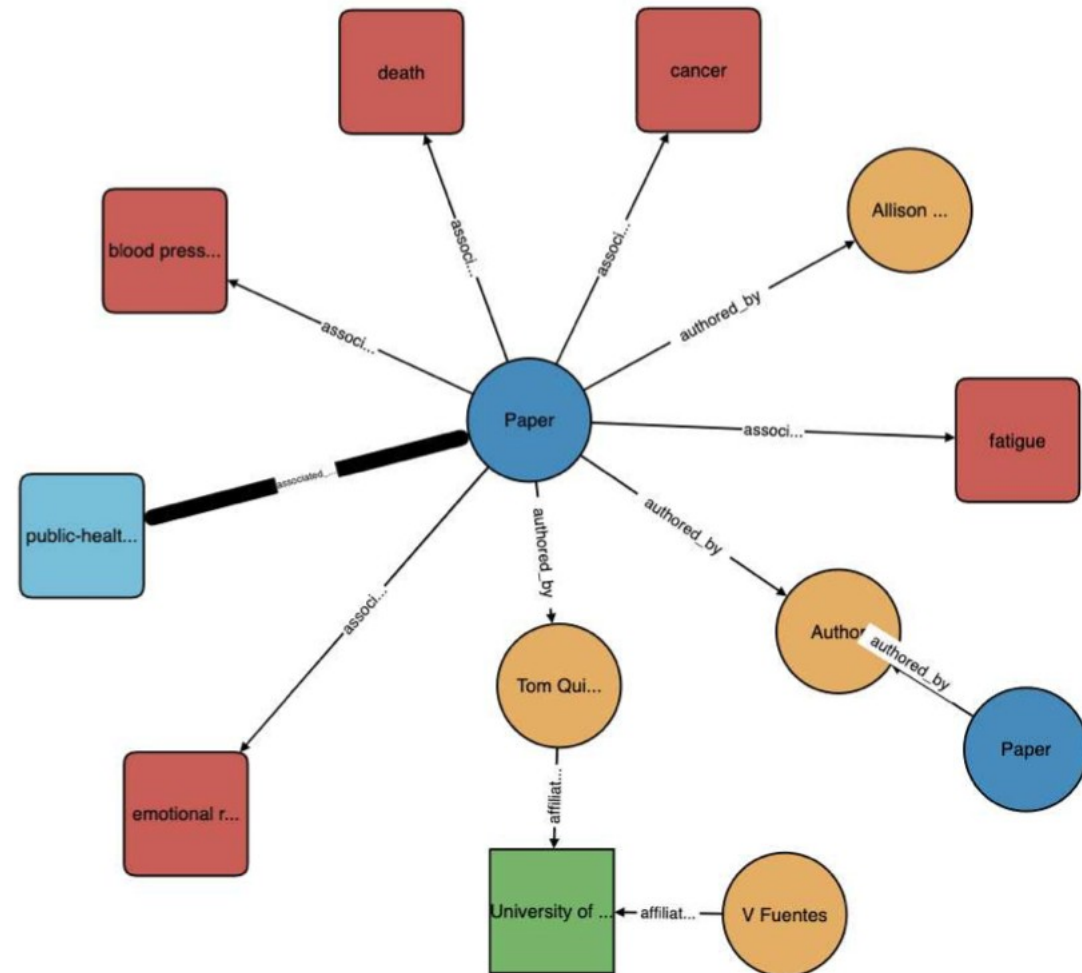


Applications

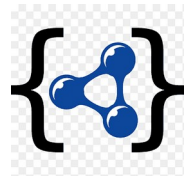
Document similarity engine that combines both semantic and relationship information from CKG from related Covid-19 research articles.

Entity Type	Count	Relation Type	Count
Papers	42,220	authored_by	240,624
Authors	162,928	affiliated_with	121,257
Institutions	21,979	associated_concept	2,739,665
Concepts	109,750	associated_topic.	95,659
Topics	10	cites	134,945
Total	336,887		3,332,151

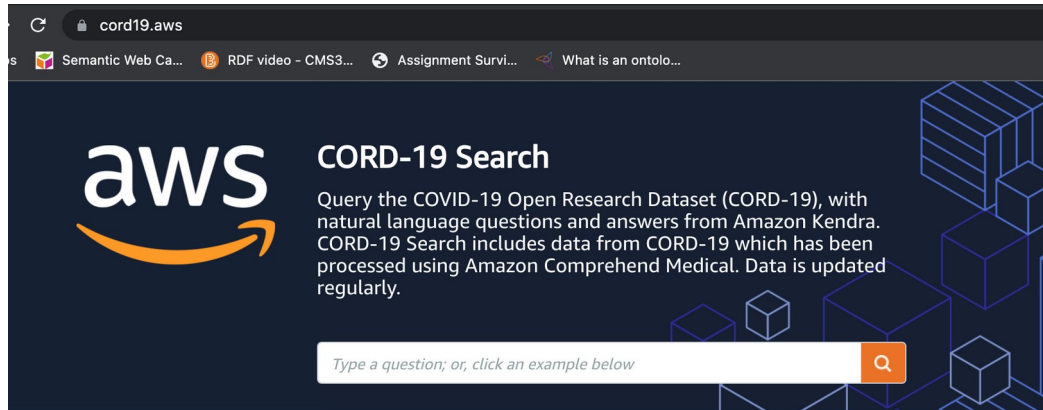
Visualization of CKG. Paper entities (blue) connect to concepts (red), topics (light blue), and authors (gold) through directed relations. Authors connect to institutions (green).



CKG Web Search Interface



Applications



Example questions

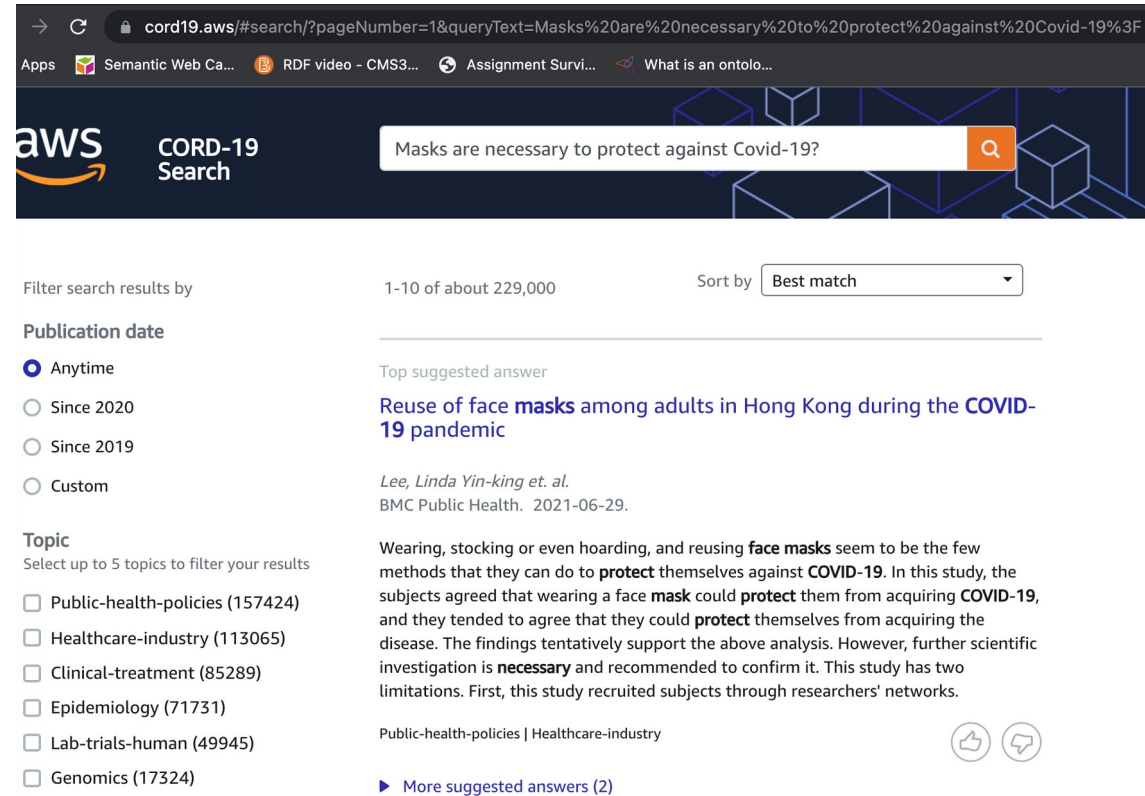
What are COVID-19 symptoms?

How infectious is COVID-19?

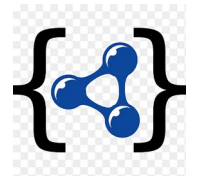
What is the mean value of R0 for COVID-19?

When is the salivary viral load highest for COVID-19?

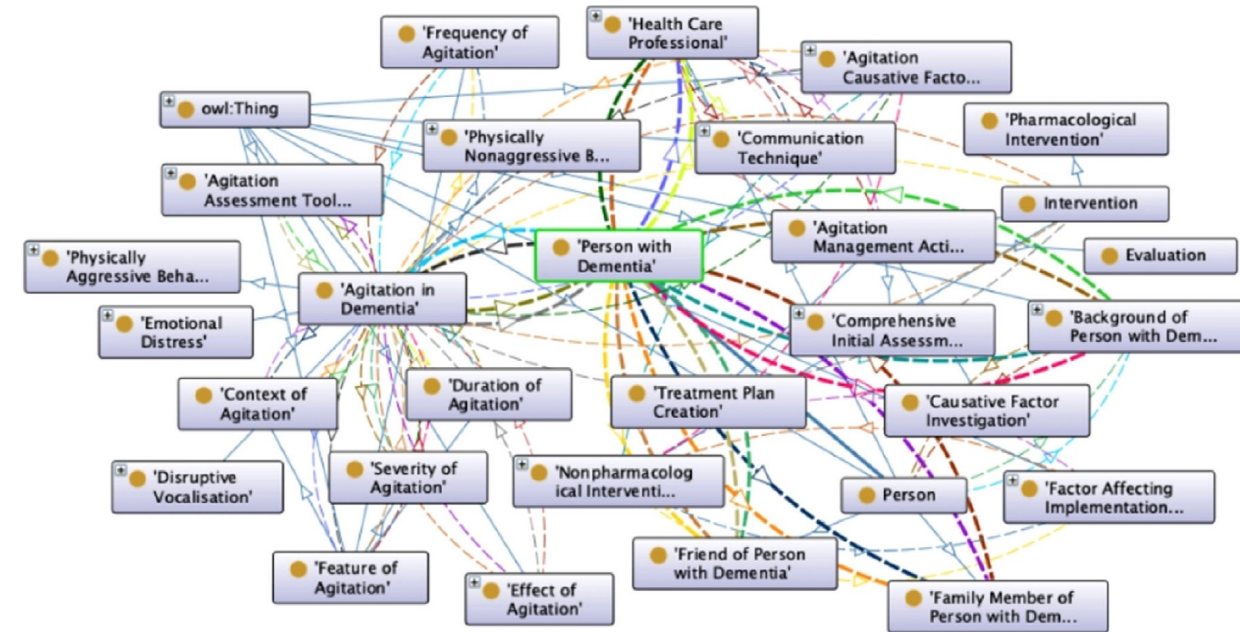
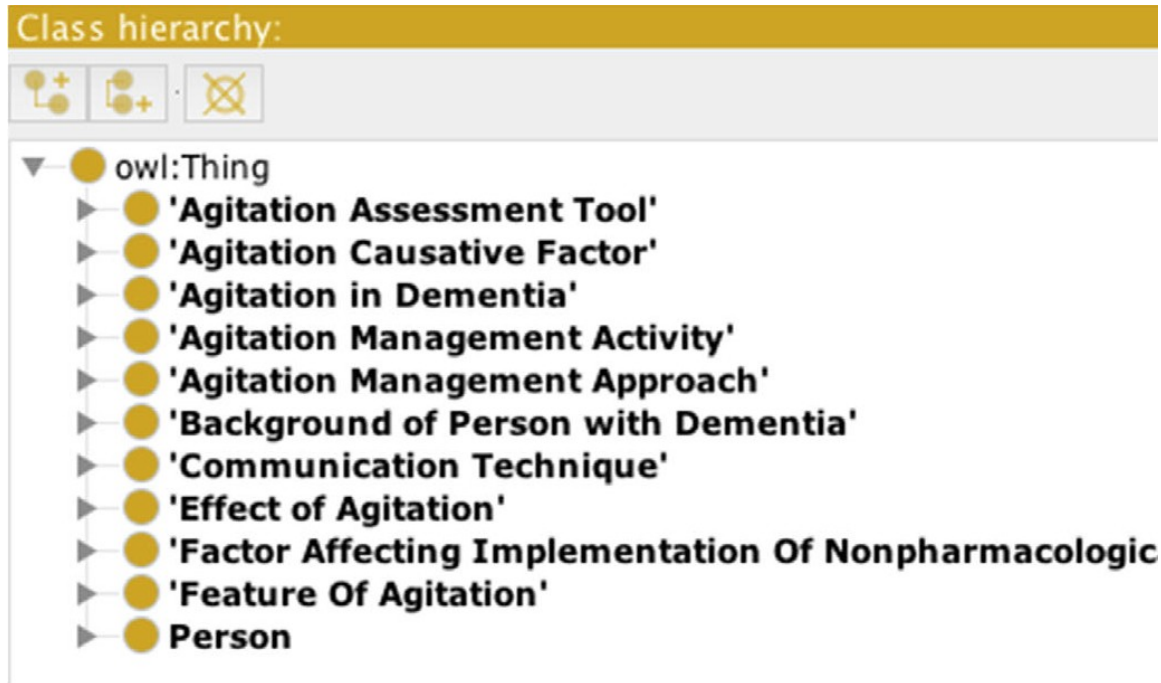
What are prevalent antibodies in saliva?



DRANPTO – Dementia Agitation Ontology

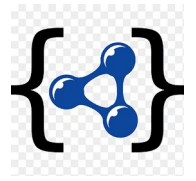


Applications



DRANPTO has 11 top classes and further 558 sub-classes.

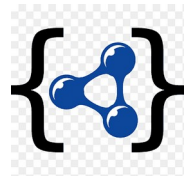
Challenges - DRANPTO



Applications

Knowledge Base	Manual Process	Ontology Evaluation	Expert-based Evaluation
• No standardized knowledge base • PRISMA based approach to gather information	• Automatic process not adopted • Non-taxonomic relationships are not captured in automatic process	• Multiple methods e.g. gold-standard based, domain expert-based & CQ-based	• Un-availability of qualified professionals • time consuming

Comparison: Automated Vs Manual Ontology Creation



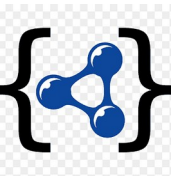
Applications

CKG is created through automatically through ML while DRANPTO is manual ontology. Purpose of CKG is research while DRANPTO is deployed in a clinical setup. However, there are few interesting comparisons can be made, although not a direct link between them.

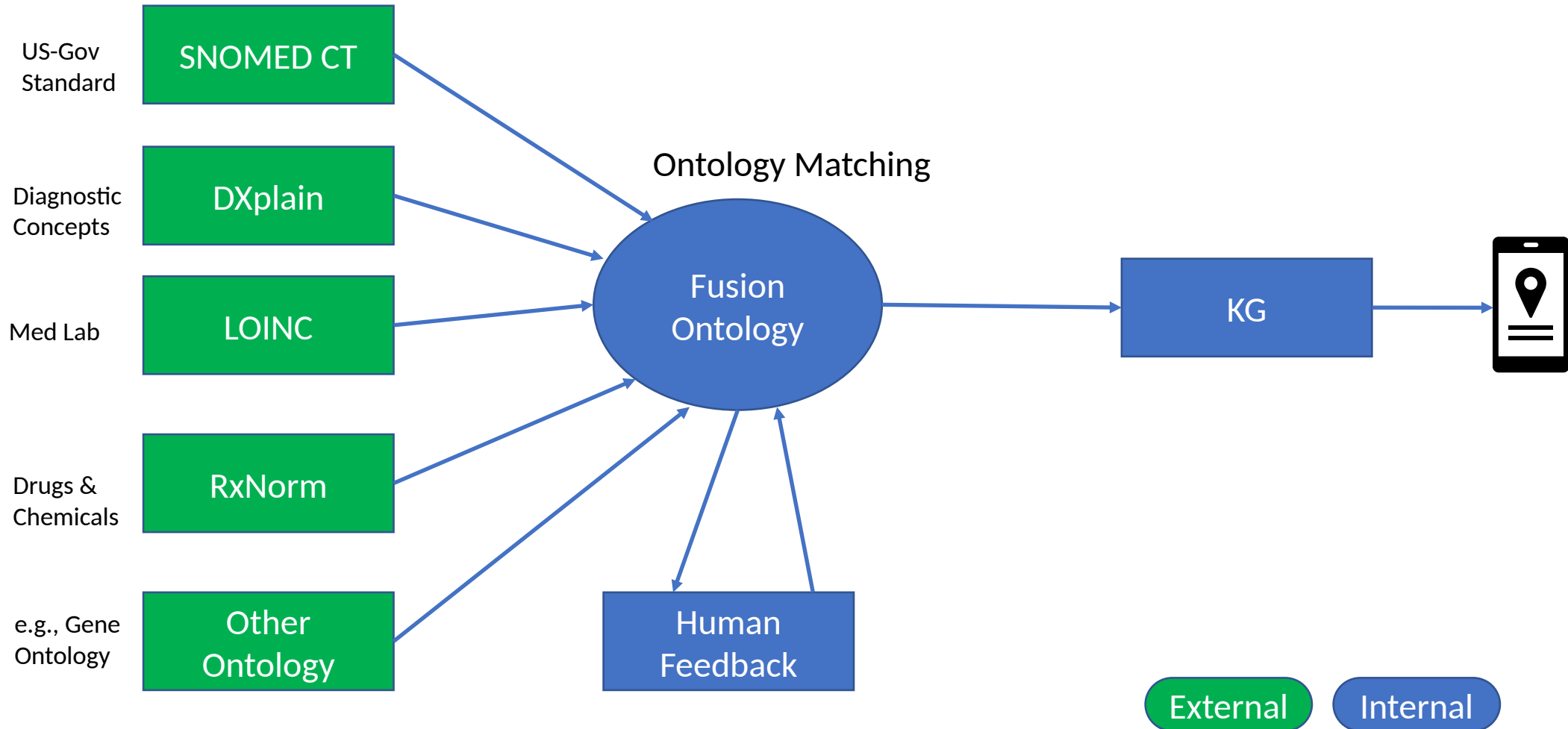
Automated Ontology Creation	Manual Ontology Creation
Non-taxonomic relations are not captured	Non-taxonomic relations can be created
Time to create ontologies is less	Longer time to complete ontology
On-going maintenance is supported by Machine Learning	Lack of domain experts poses a challenge in upgradation of the ontologies

Based on above points, we can propose that there should be semi-automatic ontology creation process which considers best of both worlds.

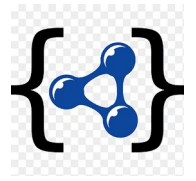
Challenges – Matching Ontologies



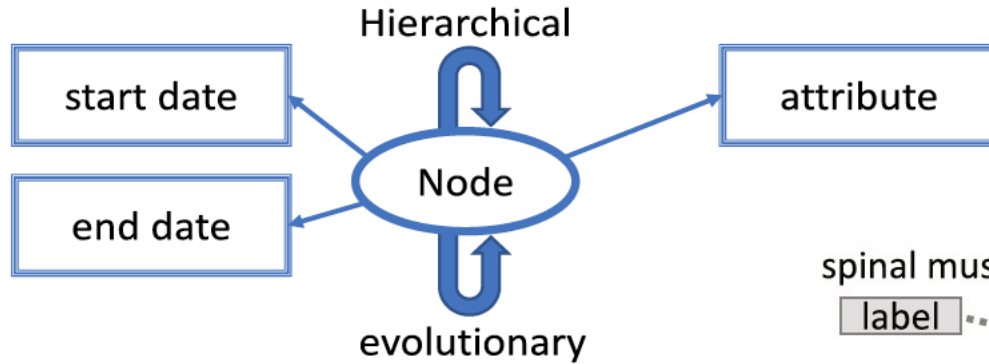
Applications



Challenges - Ontology Versions

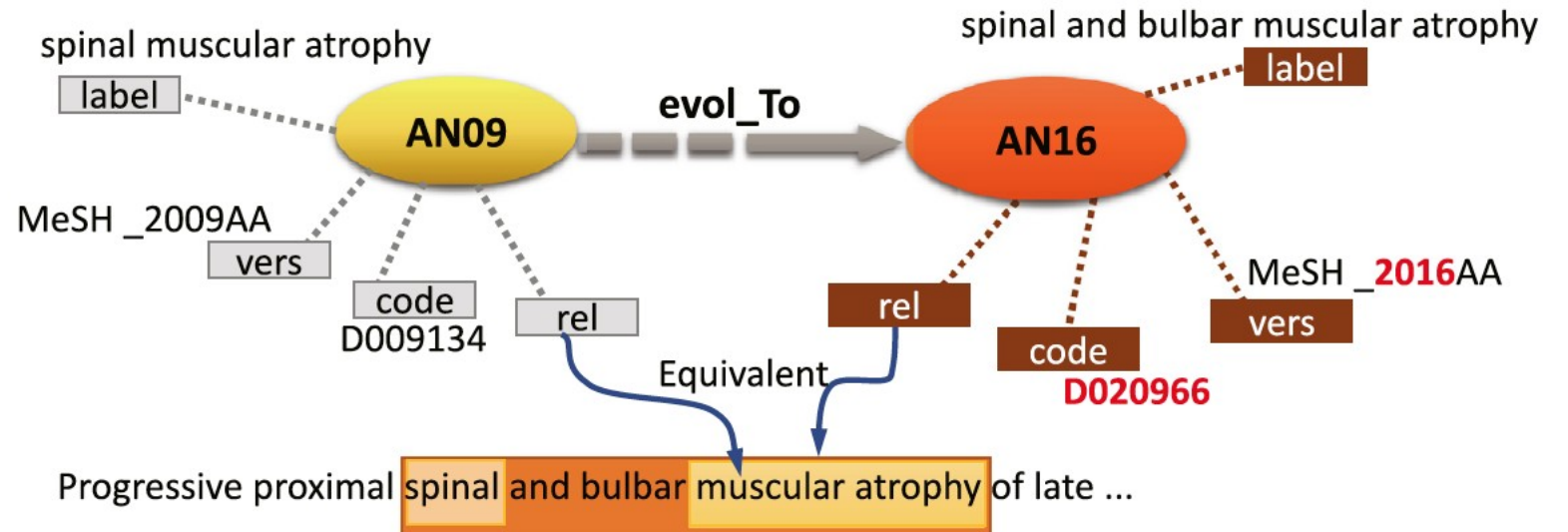


Applications



Historical KG Improves:

- Information retrieval
- maintenance of semantic annotations





Way forward
Healthcare Future

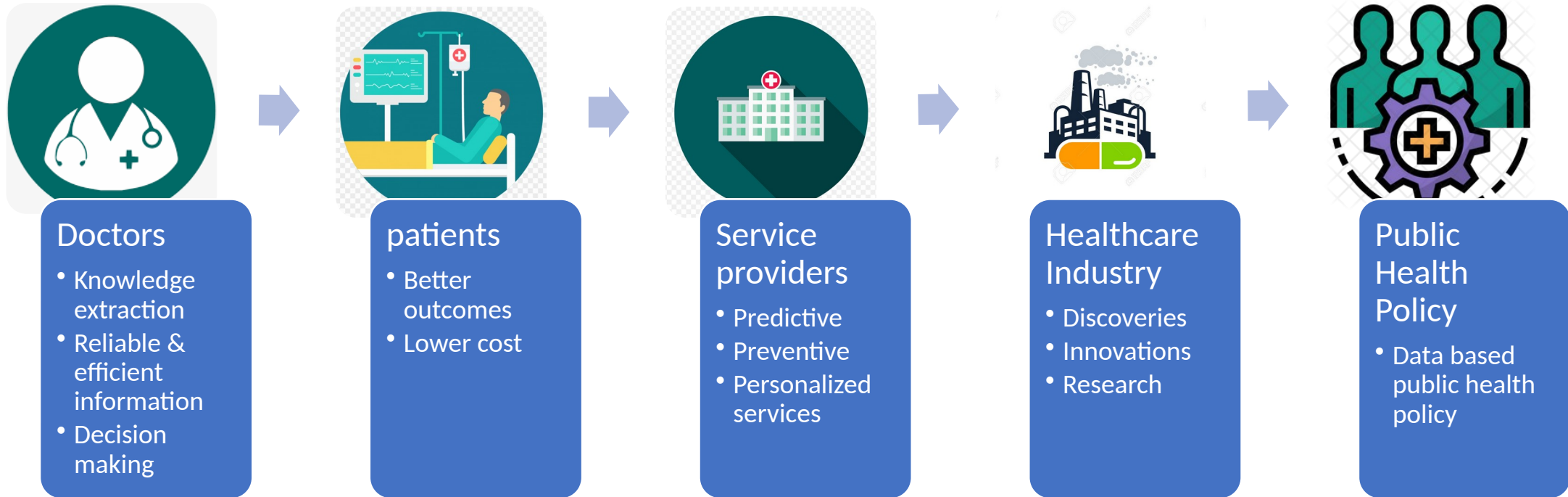
Benefits ^ Way Forward

KG for Healthcare

Benefits of KG in Healthcare



Way forward

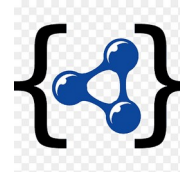
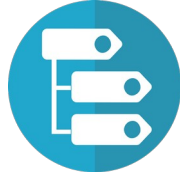




Way forward

Way Forward – KG in Healthcare

- Major Challenges
 - Upgrade ontologies due to constant innovation
 - Modulation of space & time is not properly mapped so far across Knowledge Base
- Future of KG in healthcare
 - Helpful in prediction ^ personalization of healthcare services
 - Pervasive healthcare systems
 - Smart healthcare technologies enabled by ontologies
 - Healthcare 4.0 enabled by IOT



Conclusion



Creating KGs

Knowledge
acquisition &
integration

Through
ontology,
knowledge can be
organized



Maintaining KGs

Storing &
accessing

Physical aspects
of maintaining
graphs



Exploiting KGs

Understand &
consume
knowledge

Through
enhanced search,
summarization
and answering
questions

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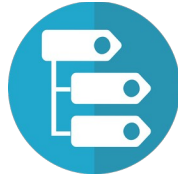


Thank you



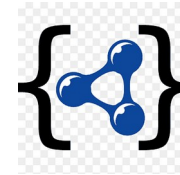
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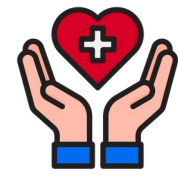
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Benefits & Way forward

Healthcare Future